

Worksheet — 1.1 Processors, Input, Output & Storage

OCR A Level Computer Science (H446) — Component 01, Section 1.1 (1.1.1 processor, 1.1.2 processor types, 1.1.3 I/O & storage).

Name: ___ Date: _____

Time: ~30 min. Write your answers in the spaces; check against the answer key at the end.

1. Key-term match

Match each term (1–8) to its definition (A–H). Write the letter on the line.

Terms

1. Register ____
2. Cache ____
3. Pipelining ____
4. Core ____
5. Volatile ____
6. Virtual memory ____
7. GPU ____
8. Control Unit ____

Definitions

- **A.** Overlapping the fetch, decode and execute of consecutive instructions to raise throughput.
 - **B.** Loses its contents when power is removed.
 - **C.** Small, very fast store inside the CPU that holds data during processing.
 - **D.** Many-core processor that performs the same operation on large data sets in parallel.
 - **E.** Use of secondary storage as extra RAM when physical RAM is full.
 - **F.** An independent processing unit able to run its own fetch–decode–execute cycle.
 - **G.** Small, fast memory on or near the CPU holding frequently used data/instructions.
 - **H.** Decodes instructions, generates control signals and synchronises the CPU with the clock.
-

2. Fill in the blanks — the Fetch–Decode–Execute cycle

Complete the passage using the word bank. Each word is used once.

Word bank: opcode · operand · PC · MDR · CIR · data · control · increment · ALU · address

During **fetch**, the address of the next instruction is copied from the ___ into the **MAR**. A read signal is sent on the _ *bus*, and the instruction returns along the _ bus into the ___. The processor then _s the PC so it points to the next instruction, and the fetched instruction is copied from the MDR into the _.

During **decode**, the Control Unit splits the instruction into the ____ (**the operation to perform**) and the ____ (**the data or** ____ to act on).

During **execute**, the instruction is carried out — for example a calculation is performed by the ____ with the result stored in the accumulator.

3. State the register

For each description, name the register (PC, ACC, MAR, MDR or CIR).

- a) Holds the address of the **next** instruction to be fetched.
- b) Holds the **address** of the location currently being read from or written to.
- c) Holds the **data or instruction** just fetched from (or about to be written to) memory.
- d) Holds the **current instruction** while it is being decoded and executed.
- e) Stores the **immediate result** of an ALU calculation.

4. Put the FDE steps in order

Number these fetch-stage steps 1–4 in the order they happen.

- _____ Read signal placed on the control bus; instruction returns via the data bus into the MDR.
- _____ The contents of the MDR are copied into the CIR.
- _____ The address in the PC is copied into the MAR.
- _____ The PC is incremented ($PC := PC + 1$).

5. Short answer — performance & architectures

- a) State **two** factors that affect CPU performance and explain the effect of each.
.....
.....
.....
- b) "Adding more cores always makes a computer faster." Give the **caveat** that makes this statement only partly true.
.....
.....
- c) Explain the key difference between **Von Neumann** and **Harvard** architectures, and give one typical use of each.
.....
.....
.....
- d) Give **two** differences between **CISC** and **RISC** processors.

e) Explain why a GPU is well suited to graphics and machine-learning work but poor at branch-heavy sequential tasks.

6. Storage device selection table

Complete the empty cells. For each storage type, state **how it works**, **one pro**, **one con** and **one example use**.

Type	How it works	One pro	One con	Example use
Magnetic (e.g. HDD, tape)				
Flash/SSD (e.g. SSD, USB, SD)				
Optical (e.g. CD, DVD, Blu-ray)				

7. Challenge box

Challenge — virtual memory & thrashing. A user keeps many large applications open and the computer becomes very slow, with the hard disk constantly active.

(i) Explain what **virtual memory** is and why the OS is using it here.

(ii) Name the problem causing the slowdown and explain **why** moving pages between RAM and disk is so slow.

(iii) Suggest **one** hardware change that would reduce the problem, and justify it.

